# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

# BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Biology (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - I) (Session 2017-2019 to 2019-2021)

**Time Allowed:** 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i Write two effects of carbon monoxide on our circulatory system.

ii How sound is produced by voice box in human?

iii Define epiglottis

iv Write two functions of spinal cord.

v What is acromegaly?

vi What are the functions of urinary bladder and urethra in urinary system of human?

vii What is homeostasis in plants?

viii How level of calcium ions is regulated in our blood?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i Differentiate between compact bone and spongy bone.

ii Define tendons. Write their functions.

iii What is meant by fragmentation and give an example.

iv Write down two main causes for spread of AIDS.

v What are nucleosomes?

vi Differentiate between transcription and translation.

vii Define loci.

viii What is difference between dominant and recessive allele?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i Define ecosystem.

ii What is biosphere?

iii What is food web?

iv Define symbiosis?

v What is biotechnology?

vi Describe the services of Pasteur for fermentation.

vii What is vaccination?

viii What is bactericidal antibiotics?

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Describe the osmoregulatory function of kidney.

Q5.(b). (Marks 5) What is cartilage? Explain three types of cartilage.

Q6.(a). (Marks 4) Define feedback mechanism. Also explain negative feedback and positive feedback.

Q6.(b). (Marks 5) Compare adaptations in insect-pollinated and wind-pollinated flowers.

Q7.(a). (Marks 4) Write a note on medicinal drugs.

Q7.(b). (Marks 5) Write a note on applications of fermentation.

## GROUP – II

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## Biology (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - II) (Session 2017-2019 to 2019-2021)

| **Time Allowed:** 15 Min. | **Maximum Marks:** 12 |
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> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which hormone is related to child birth :

(A) Oxytocin
(B) Vasopressin
(C) ADH
(D) Thyroxin

Q2. Which is a biotic component of an ecosystem :

(A) Air
(B) Water
(C) Soil
(D) Producers

Q3. If an organism has two different alleles for a single trait, its genotype is said to be :

(A) Homozygous
(B) Hemizygous
(C) Heterozygous
(D) Homologous

Q4. Extra water is removed from the plant body by :

(A) Respiration
(B) Transpiration
(C) Guttation
(D) Evaporation

Q5. The number of chromosomes in human body cells are:

(A) 42
(B) 44
(C) 46
(D) 48

Q6. Which drug is obtained from animals :

(A) Fish liver oil
(B) Digitalis
(C) Aspirin
(D) Morphine

Q7. The largest part of forebrain is :

(A) Thalamus
(B) Hypothalamus
(C) Cerebrum
(D) Cerebellum

Q8. The joints between skull bones are called :

(A) Moveable
(B) Immoveable
(C) Slightly moveable
(D) Hinge joints

Q9. ----------seed have epigeal germination :

(A) Maize
(B) Coconut
(C) Pea
(D) Cotton

Q10. The hormone which may prove effective against brain and lung cancer is called :

(A) Insulin
(B) Thymosin
(C) Interferons
(D) Urokinase

Q11. Many other chemicals in smoke increase production of ------ blood cells :

(A) Red
(B) White
(C) Basinophil (printed as "Basinophil" in the source scan)
(D) Platelets

Q12. A recycling of one tonne of paper can save trees :

(A) 17
(B) 18
(C) 19
(D) 20

## Biology (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - II) (Session 2017-2019 to 2019-2021)

**Time Allowed:** 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i Differentiate between breathing and cellular respiration.

ii Define pneumonia.

iii Write the effect of smoking on circulatory system.

iv Differentiate between renal cortex and renal medulla.

v What is tubular secretion?

vi What is ganglion?

vii What is meant by brain stem?

viii Write any two functions of spinal cord.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i Define skeleton.

ii What are osteocytes?

iii Define reproduction.

iv Define parthenogenesis.

v Define inheritance.

vi What is transcription?

vii Write down characteristics of pea plant.

viii What is punnett square?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i What is meant by ecosystem? Give two examples of natural ecosystem.

ii Differentiate between population and community.

iii Define food chain and give an example.

iv Write the symptoms of dengue fever.

v Define biotechnology.

vi Write the uses of ethanol.

vii What are synthetic drugs? Give an example.

viii Define antibiotics and give an example.

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Describe the osmoregulatory function of kidney.

Q5.(b). (Marks 5) Describe the components of human skeleton.

Q6.(a). (Marks 4) Describe the structure of a neuron.

Q6.(b). (Marks 5) Explain various methods of artificial vegetative propagation in plants.

Q7.(a). (Marks 4) Describe four sources of medicinal drugs giving examples.

Q7.(b). (Marks 5) Define fermentation and describe its industrial products.